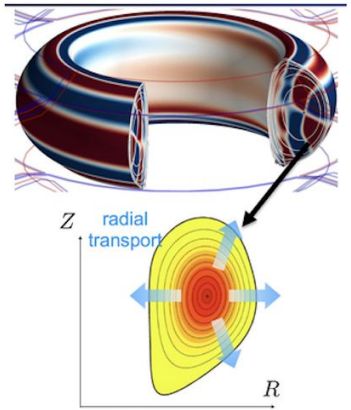
DSC 180A Week 5 Check In

Group D10 A: Fusion Reactor Simulation Surrogate Models at General Atomics

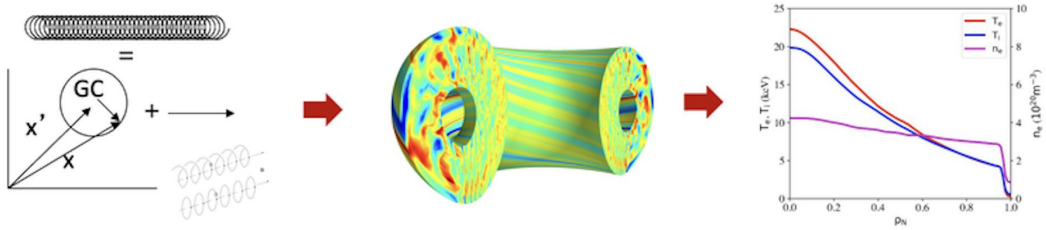
Sweekrit Bhatnagar, Aryan Nath, Ryan Lindberg, Joseph Calderon
Mentors: Igor Sfiligoi, Jeff Candy (GA) , Emily Belli (GA)

Project Overview

- Develop a machine learning model that approximates TGLF predictions of turbulent transport in fusion plasmas.
    - TGLF = quasi-linear model to solve fluid-like equations inside tokamaks (reactors that use strong magnetic fields to accelerate particles and cause fusion)
- Predict ion heat flux, electron heat flux, and particle flux from plasma parameters.
- Reduce computational cost while preserving predictive accuracy.



Tokamak geometry and radial particle/energy transport



Particle gyration -> turbulence -> radial plasma profiles

Project Goals

- Build a fast, accurate surrogate for TGLF
  - This was previously achieved with TGLF-NN
- Compare different machine learning approaches.
- Evaluate model accuracy, computational efficiency, and generalization
- Recreate and investigate current literature + increase speed of training/inference

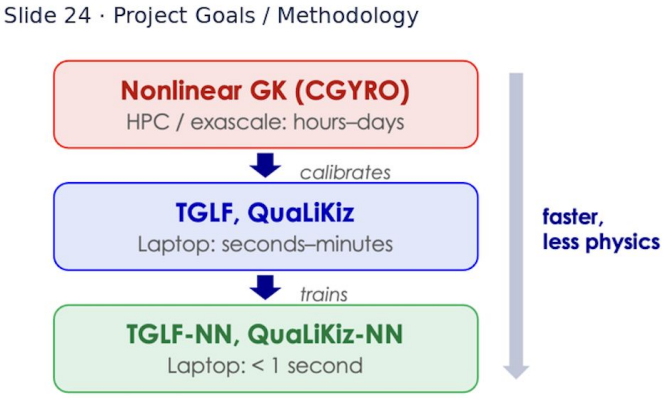
1A Neural-net surrogate for TGLF

Goal: learn TGLF (~15 inputs → fluxes) in milliseconds

Data: TGLF database, inputs derived from exp

- Find correlations among input parameters
- Compare regression, trees, and neural nets

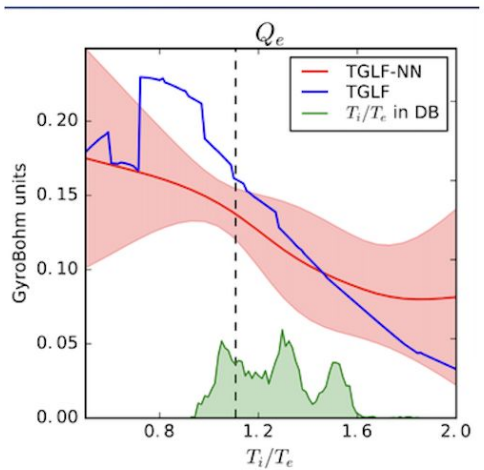
Emily Belli's exact Project 1A brief (~15 inputs -> fluxes)



How nonlinear CGYRO calibrates TGLF, and TGLF trains an NN surrogate

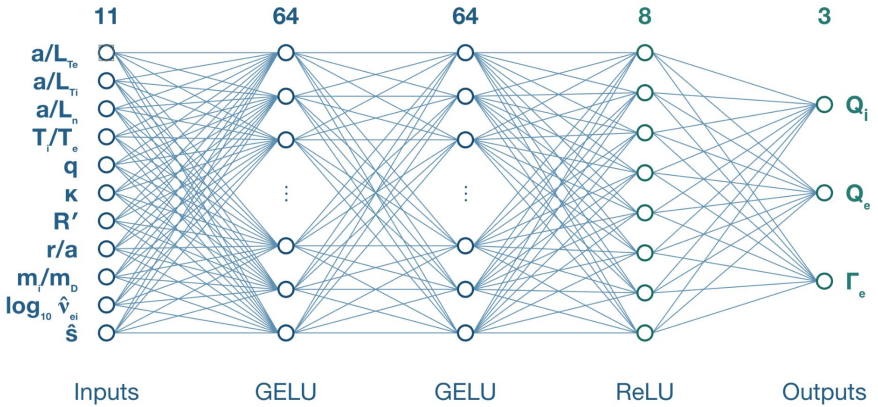
Methodology

- Obtain and preprocess simulation data generated by TGLF.
- Train and compare regression models, tree-based methods, neural networks, and neural network ensembles.
- Evaluate performance using prediction errors, inference speed, and behavior near instability thresholds.



Example TGLF-NN vs TGLF curves and surrogate smoothing

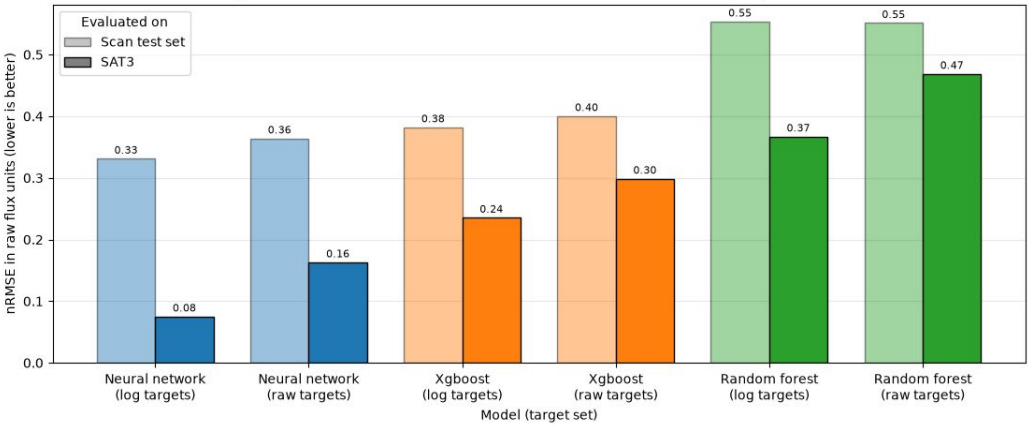
Five-network ensemble



Progress and Results

- Reviewed the underlying physics and differences between TGLF and nonlinear CGYRO simulations. (Weeks 1-2)
- Identified potential modeling approaches and evaluation metrics (Week 3)
  - Leaning towards ensemble neural network methods
- Currently establishing dataset availability, baseline models, and experimental requirements
  - Received access to TGLF simulation to create high-dimensional datasets

Figure 3. Error on the scan test set vs. SAT3 for each model and target set





Team Contributions

Ryan: Tested NN ensemble methods for predicting TGLF fluxes, inspired by current literature, indicating promising results to explore going forward.

Sweekrit: Thoroughly tested neural nets, random forests, and gradient boosted decision trees, helping us rule out which methodologies were useful or not going forward.

Aryan: Tested Histogram-Based Gradient Boosting Regressors, which showed promising results, and fast compute times. A potential avenue of exploration in addition to NN ensemble methods.

Joey: Built upon Ryan’s and Sweekrit’s models for neural networks to provide further improvements and better replicate TGLF-NN results.

Weekly Contributions

Weeks 1-2: Each group member individually read current literature and research papers assigned by our mentors. This period was meant to build a knowledge foundation of the problem.

Week 3: Each group member individually explored a small TGLF dataset. Then we created a presentation as a group to discuss and show results during our meeting. This was meant to build a stronger understanding of the data we would be working with.

Week 4: Each group member individually experimented with an expanded dataset of 5000 cases, testing out different modeling techniques, ruling out bad results and uncovering promising methodologies for moving forward. Each group member found promising results that helped narrow the scope of our project and gain an understanding of current modeling techniques.